

Telemetry Packet Evidence

Verify bytes before trusting an observation

A local fixture makes integrity, sequence, and age visible.

Lesson	026 — component	Time	150–210 minutes
Prerequisites	Lessons 001–025	Board	Arduino Mega 2560
Primary evidence	D45 / TP-LED	Status	host verified; bench open
Safety boundary	E1, compiled-in fixtures, no radio receiver or transmitter		

1 Mission

Preserve the evidence chain:

packet bytes → integrity → accepted observation → local age → visible cadence.

`TelemetryPacketCodec` defines one exact bounded ADK lab packet. `ObservationTracker` makes sequence and local freshness explicit. This packet is not an observed radio protocol. The canonical sketch creates its fixtures locally and neither receives nor emits RF.

Integrity is not trust. CRC-16 detects accidental changes. It does not authenticate a sender, hide a payload, prove provenance, prevent replay, or make an unknown capture lawful to retain.

1.1 Learning outcomes

- distinguish packet, sample, observation, and record;
- explain exact-length decoding and unchanged output on failure;
- classify first, in-order, duplicate, gap, and reordered sequence values;
- calculate freshness from local receipt time rather than a remote clock;
- separate transport corruption, sample quality, and staleness;
- prove acquisition and safe state with separate circuit evidence.

2 Parts, energy, and wiring

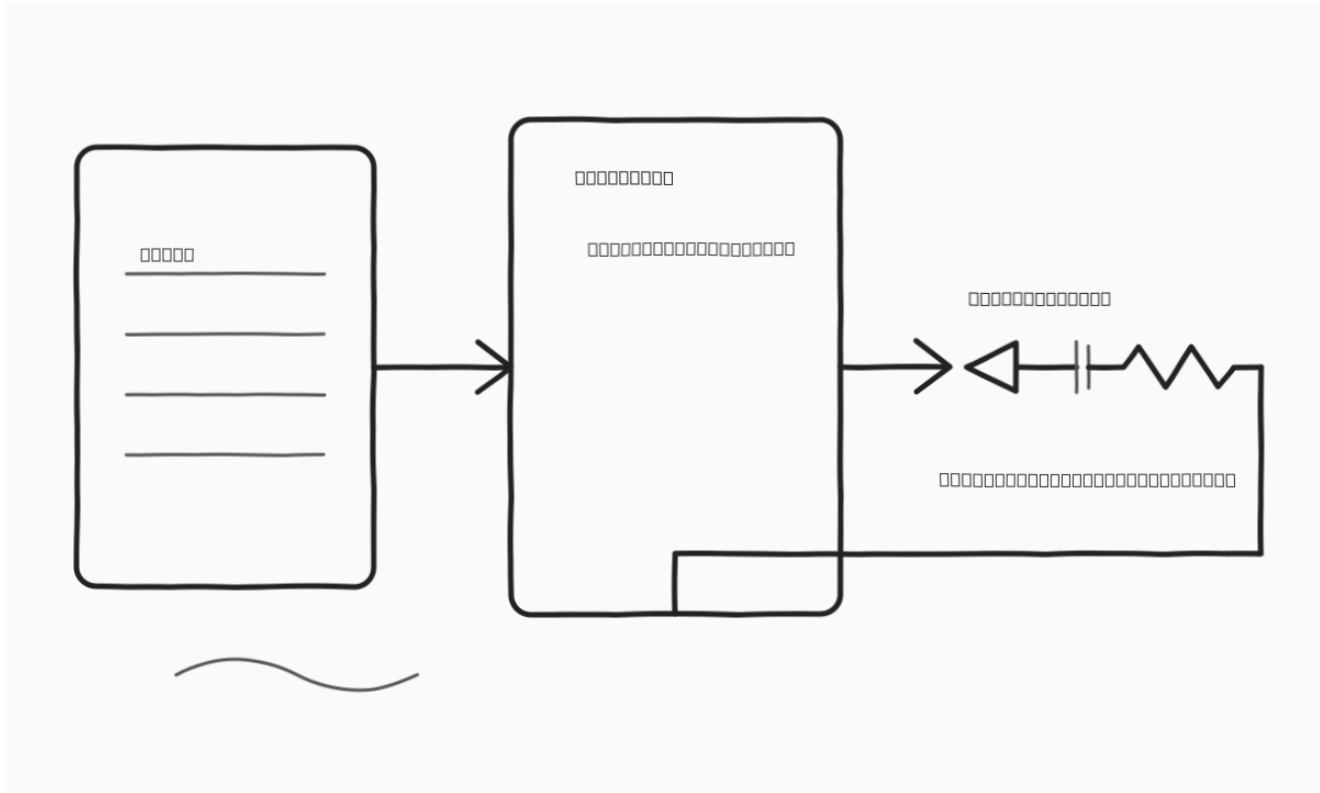
Part	Requirement
Arduino Mega 2560	USB-powered board
Evidence LED	One ordinary LED and one 1 kΩ resistor
Instrument	Logic probe or oscilloscope; meter for static safe-state check
Fixtures	Compiled-in ADK-owned bytes; no external receiver

For an illustrative 5 V rail and 2 V LED drop,

$$I = \frac{5 \text{ V} - 2 \text{ V}}{1000 \text{ } \Omega} = 3 \text{ mA}.$$

Measure the actual rail, resistor, and LED drop before recording current. Remove USB power before changing wires or probe clips.

Signal	Mega pin	Connection
Acquisition evidence	D13	built-in LED
Observation evidence / TP-LED	D45	D45 $\rightarrow$ 1 k Ω $\rightarrow$ LED anode; cathode $\rightarrow$ GND
Probe reference	GND	common reference for TP-LED



Alternative text: graphite orientation plate showing local valid, gap, corrupt, and silent fixture packets entering a Mega; the Mega checks integrity, sequence, and age, then drives one resistor-limited D45 evidence LED. No receiver, antenna, or transmitter is present.

3 Exact bounded packet

The wire image is 19 bytes. It contains no pointer, padding, native enumeration, native Boolean, or compiler-dependent structure image. Multibyte fields are big-endian; signed fields use two's-complement bits.

Offset	Bytes	Field	Meaning
0	1	magic	0xAD
1	1	version	0x01
2	2	source ID	opaque lab identifier
4	2	sequence	ordering evidence
6	4	observed ms	untrusted remote-clock evidence
10	1	kind	bounded telemetry kind
11	1	quality	bounded sample quality
12	4	value	signed integer
16	1	exponent	signed decimal scale
17	2	CRC-16	accidental-corruption check

CRC-16/CCITT-FALSE uses polynomial 0x1021, initial value 0xFFFF, no reflection, no final XOR, and covers bytes 0–16. It is not a security primitive.

`encode()` returns `Result<uint16_t>`; callers check `ok()` before reading `value()`. A short output span receives no partial packet. `decode()` requires exactly 19 bytes and distinguishes bad version, length, type, quality, integrity, and trailing data. Failure leaves the destination unchanged.

3.1 Packet worksheet

Check	Prediction	Observation
Golden valid packet	accepted	_____
One changed value bit	bad integrity	_____
One byte removed	bad length	_____
One byte appended	trailing data	_____
Unknown kind	bad type	_____
Unknown quality	bad quality	_____

A changed value or timestamp bit with the old CRC reaches **BadIntegrity**. A changed magic, version, kind, or quality may reach its semantic validator first. Tests recompute CRC after a deliberate semantic change when isolating **BadVersion**, **BadType**, or **BadQuality**. Every one-bit mutation is rejected; the exact class follows this precedence.

4 Sequence is evidence, not arrival order

The tracker owns one source ID and remembers one accepted observation. **First** establishes the baseline. The next sequence is **InOrder**; a larger forward jump is **Gap**. Equal is **Duplicate**; evidence behind the accepted value is **Reordered**. Comparison uses a documented unsigned half-range rule so wrap from 65535 to 0 remains deterministic.

The exact delta 0x8000 is **Reordered**; it never replaces the accepted observation.

Duplicate and reordered packets do not replace the accepted sample. A forward gap may replace it, but the gap remains visible. Source mismatch is invalid input rather than an observation for a different tracker.

Accepted	Incoming	State	Does value change?
—	100	First	yes
100	101	InOrder	yes
101	101	Duplicate	no
101	104	Gap	yes, with gap evidence
104	103	Reordered	no
65535	0	InOrder	yes, across wrap

Half-range prediction: accepted _____; incoming _____; expected classification _____; reason _____.

5 Freshness belongs to the receiver

Age is `now - receivedAt`, not `now - observedMilliseconds`. The remote timestamp is retained for diagnosis, but a future, reversed, or drifting remote clock cannot keep a packet fresh.

The example uses:

$$\text{Fresh} < 2000 \text{ ms}, \quad 2000 \leq \text{Aging} < 5000 \text{ ms}, \quad \text{Stale} \geq 5000 \text{ ms}.$$

Local age	Prediction	Observation
1999 ms	Fresh	_____
2000 ms	Aging	_____
4999 ms	Aging	_____
5000 ms	Stale	_____

Now reverse the packet's remote timestamp while keeping the same local receipt time.

Prediction: _____.

Interpretation: _____.

6 Narrative run loop and visible state

```
observePacket      (now);
verifyPacket      (now);
updateFreshness   (now);
showObservationEvidence (now);
```

Objects appear in dependency order: runtime, acquisition LED, packet-evidence LED, pure codec, then tracker. Setup acquires, configures, and starts. The loop observes one scheduled fixture, verifies exact packet bytes, updates freshness, and actuates D45.

The 12-second schedule accepts sequence 1, skips 2 and accepts 3, corrupts the next packet, waits until sequence 3 becomes stale, then accepts sequence 4 after wrap. Every advertised evidence state is reachable, and recovery is an accepted in-order observation rather than a display reset.

TelemetryFixtureSchedule owns this tested one-shot decision trace; the sketch does not duplicate its timing policy. **TelemetryEvidenceModel** applies the tested precedence fault, corrupt, stale, sequence/aging, then fresh. The four run-loop calls therefore name actual work rather than forwarding into another private state machine. An unexpected execution fault uses the safe-off path; the four-pulse fixture demonstrates corrupt packet evidence, not a claim that a failed output can diagnose itself.

State	D45 cadence	Meaning
Fresh, in order	one 1 s pulse	accepted observation is fresh
Sequence or aging	two 150 ms pulses	duplicate, gap, reorder, or aging
Stale	three 120 ms pulses	no recent accepted observation
Corrupt fixture	four 100 ms pulses	packet integrity failed

D13 proves acquisition only. D45 reports current evidence only. Serial may print supporting records, but it does not control tuning, capture, or replay and is never the sole observation.

7 Predict, observe, interpret

7.1 Experiment A: acquisition

Predict. D13 illuminates only after the tracker and both output endpoints initialize.

Observe: _____. **Interpret.** This supports acquisition, not packet correctness or safe shutdown.

7.2 Experiment B: deterministic fixture cycle

Before upload, predict each D45 cadence. Probe TP-LED against GND and record:

Fixture	Predicted count	Observed count	Pulse widths
valid	_____	_____	_____
forward gap	_____	_____	_____
corrupt	_____	_____	_____
silence to stale	_____	_____	_____

Agreement supports the path from fixture bytes through integrity, tracker, and visible actuation. It proves no RF, network, remote sensor, or security claim.

7.3 Experiment C: safe state

Invoke shutdown. Predict D13 and D45 inactive and both pins released. Observe the LEDs, then measure TP-LED separately. A dark LED does not prove high impedance or claim release.

Measured safe-state evidence: _____.

8 Diagnosis and privacy review

Observation	Check next
D13 dark	tracker configuration, output claim conflict, board power
D13 on; D45 dark	fixture schedule, LED polarity, resistor, TP-LED
valid becomes corrupt	exact length, version, byte order, CRC coverage
gap never appears	sequence fixture and half-range comparison
stale never appears	local receipt time and exact 5 s boundary
state changes from remote clock	defect: freshness must use local receipt
D45 driven after stop	safe-state failure; remove USB and record deviation

8.1 Minimum-data worksheet

For every proposed field ask: Does the experiment require it? Could it identify a person, place, account, or device? Can an opaque lab identifier replace it? How long must it be retained?

Proposed field	Required?	Opaque substitute?	Retention reason
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____

Use only owned lab fixtures. Do not collect device serial numbers, addresses, locations, credentials, or unnecessary payloads. This lesson contains no receiver, transmitter, raw capture export, rolling-code analysis, key recovery, replay, or command mapping.

9 CLI and open acceptance records

```
make host-test
make arduino-Lesson026TelemetryPacket
```

```

make size-Lesson026TelemetryPacket
make upload-Lesson026TelemetryPacket PORT=/dev/ttyACM0
make serial-log PORT=/dev/ttyACM0
make lessons
make lessons-check
make site-check

```

Software fixture acceptance

- ☐ Golden valid packet and exact encoded bytes compared.
- ☐ All one-bit corruption positions rejected by host tests.
- ☐ Truncation, extension, unaligned input, and short output tested.
- ☐ Sequence first/in-order/duplicate/gap/reorder/wrap tested.
- ☐ Freshness edges and remote-clock independence tested.
- ☐ Identical fixtures and receipt times replay byte identically.

Commit _____; compiler _____; result _____.

Mega bench acceptance—open until measured

- ☐ Exact Mega, LED, resistor, supply, and instrument recorded.
- ☐ Unpowered D45 wiring and polarity checked.
- ☐ D13 acquisition evidence observed separately.
- ☐ One-, two-, three-, and four-pulse patterns measured.
- ☐ TP-LED widths and periods recorded against GND.
- ☐ Reset, shutdown, USB removal, and claim reuse observed.
- ☐ D45 safe state checked separately from LED appearance.

Supply _____; instrument _____; date _____; reviewer _____. Deviations: _____.

Current result: host verified; hardware acceptance open. Fixtures, compilation, and blank worksheets are not physical verification.

10 Exercises and further reading

1. Change each bit of one golden packet in turn. Explain why rejection establishes error detection but not sender identity.
 2. Prove that a one-byte-short encoder destination remains unchanged.
 3. Draw the five sequence classes around 16-bit wrap and half range.
 4. Make the remote timestamp move backward while local time advances. Predict freshness before running the test.
 5. Propose a minimal humidity sample record. Remove every field that the experiment does not need.
 6. Explain why sample quality `SensorFault`, transport corruption, and local staleness must remain three different observations.
- HTML reference and corrections
 - TelemetryPacketCodec public header and implementation
 - packet golden-vector and corruption tests
 - ObservationTracker public header and implementation
 - tracker lifecycle and timing tests
 - fixture schedule and evidence model and fixture and visible-precedence tests
 - Lessons 025–027 executable design
 - Arduino Mega 2560 Rev3 datasheet